

ANALYSIS

Customer Shopping Behavior Analysis



Analyzing 3,900 purchases across product categories to uncover spending patterns, customer segments, and subscription behavior for strategic business decisions.

DATASET OVERVIEW

The Data Behind the Insights

3,900

Total Purchases

18

Columns

\$59.76

Avg Purchase

3.75

Avg Rating

Key Features

- Demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item, Category, Amount, Season, Size, Color
- Behavior: Discount, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping Type

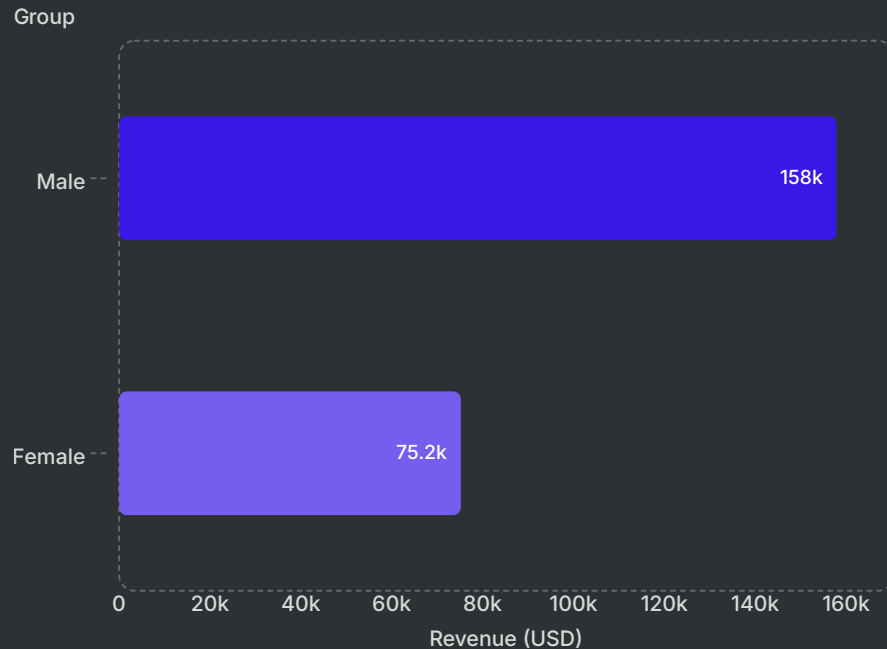
Missing data: 37 values in Review Rating, imputed using category median.

Preparing the Data

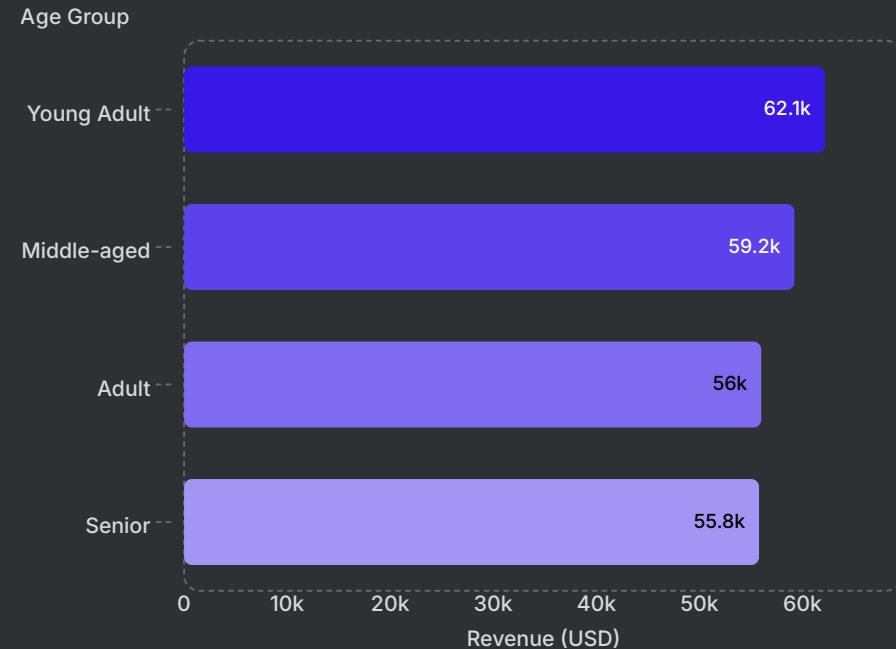
Columns standardized to snake case. Redundant `promo_code_used` column dropped after confirming it mirrored `discount_applied`.
Cleaned data loaded into PostgreSQL for structured analysis.



Revenue by Gender & Age Group



Male customers generate over **2× the revenue** of female customers.



Young Adults lead revenue contribution across all age segments.

Customer Segmentation & Subscriptions



Subscription Impact

Non-Subscribers

2,847 customers ·
\$170,436 total
revenue · \$59.87 avg
spend

Subscribers

1,053 customers ·
\$62,645 total
revenue · \$59.49 avg
spend

Repeat buyers with 5+ purchases: **2,518 non-subscribers** vs **958 subscribers**.

Top Products by Rating & Discount Dependency

Highest Rated Products

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Top Products per Category

Accessories

Jewelry (171), Sunglasses (161), Belt (161)

Clothing

Blouse (171), Pants (171), Shirt (169)

Footwear

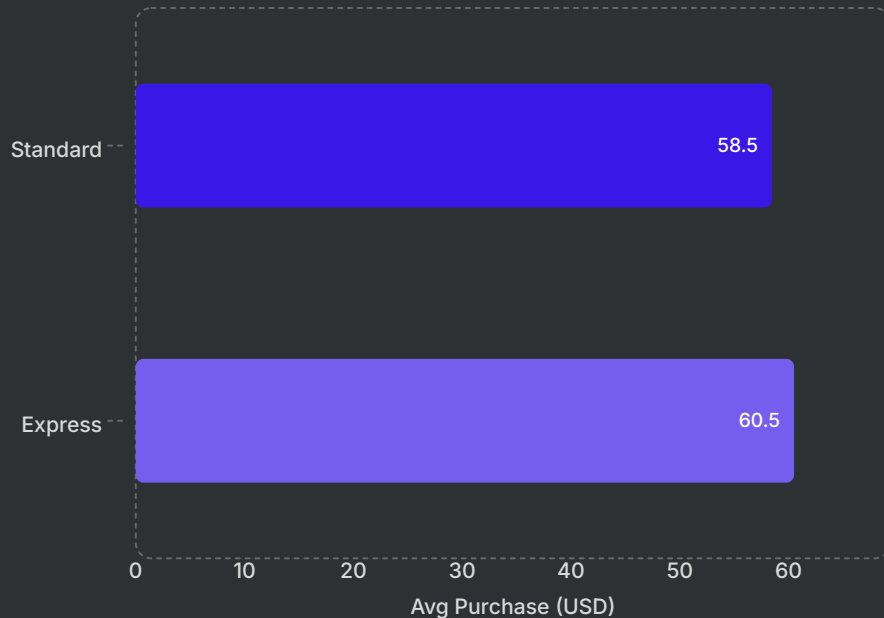
Sandals (160), Shoes (150), Sneakers (145)

Outerwear

Jacket (163), Coat (161)

Shipping & High-Spending Discount Users

Shipping Type



Discount Users Spending Above Average

839 customers used discounts yet still spent above the average purchase amount of \$59.76.

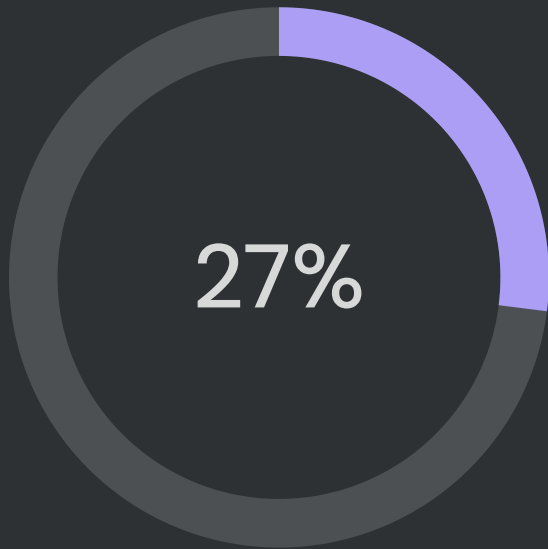
Express shipping customers spend **\$2 more** on average per purchase.

Key Insight

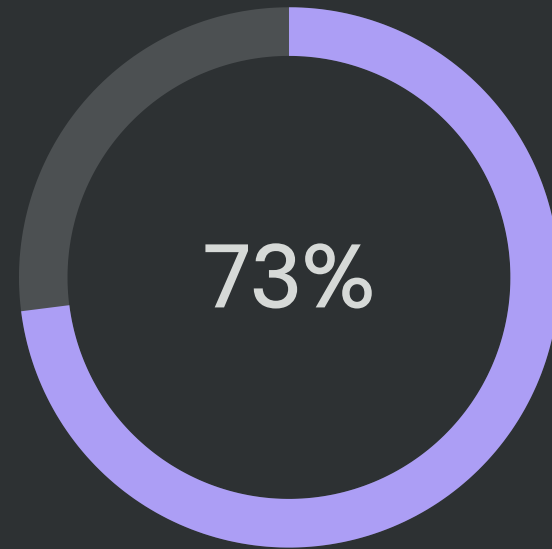
Discounts don't always mean lower spending — many discount users are high-value customers.

Visualizing the Insights

An interactive Power BI dashboard brings the analysis to life with filters for subscription status, gender, category, and shipping type — surfacing key metrics across 3,900 customers.



Subscribers



Non-Subscribers

Strategic Business Recommendations

1

Boost Subscriptions

Promote exclusive benefits for subscribers to grow the 27% base.

2

Loyalty Programs

Reward repeat buyers to move them into the Loyal segment.

3

Review Discount Policy

Balance sales boosts with margin control — 839 high-spenders use discounts.

4

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

5

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.